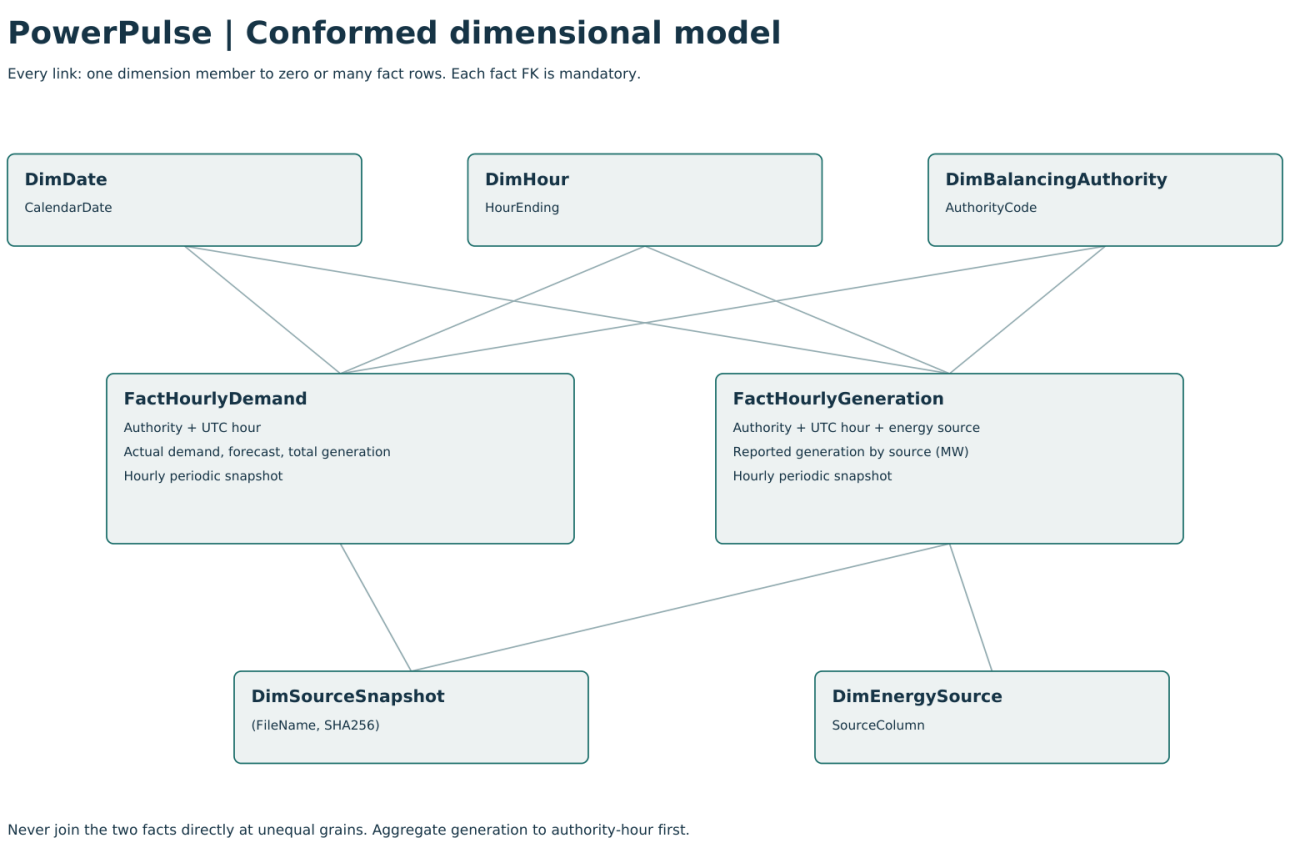
06

# Schema atlas

Conceptual, logical and physical design | PostgreSQL target




Conformed star constellation: five dimensions and two hourly periodic snapshot facts.

## Business processes and cardinality

Demand and day-ahead forecast monitoring; net generation mix monitoring. Each dimension may describe zero or many rows in its related facts. Every fact row references exactly one member of each listed dimension.

## Safe comparison

FactHourlyDemand is unique on authority and UTC hour. FactHourlyGeneration adds energy source to that grain. Aggregate generation to authority and UTC hour before joining to demand. For a group comparison, aggregate each fact independently to the same dimensions and then combine. One shared authority-hour with eight fuel categories still has one demand observation.

06

# DimDate

Logical and physical specification | Conformed dimension

**Grain: One local operating calendar date.**

| Column | PostgreSQL type | Key / nullability |
|---|---|---|
| **DateKey** | bigint | PK / NOT NULL |
| **CalendarDate** | date | UQ / NOT NULL |
| **CalendarYear** | smallint | Attribute / NOT NULL |
| **QuarterNumber** | smallint | Attribute / NOT NULL |
| **MonthNumber** | smallint | Attribute / NOT NULL |
| **ISOWeekYear** | smallint | Attribute / NOT NULL |
| **ISOWeekNumber** | smallint | Attribute / NOT NULL |
| **WeekdayNumber** | smallint | Attribute / NOT NULL |
| **Season** | text | Attribute / NOT NULL |

**Unique natural/business key: CalendarDate.**

History / update policy: Type 0. Calendar attributes are deterministic.

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# DimHour

Logical and physical specification | Conformed dimension

**Grain: One local clock hour ending label, 00 through 23.**

| Column | PostgreSQL type | Key / nullability |
|---|---|---|
| **HourKey** | bigint | PK / NOT NULL |
| **HourEnding** | smallint | UQ / NOT NULL |
| **HourLabel** | text | Attribute / NOT NULL |

**Unique natural/business key: HourEnding.**

History / update policy: Type 0. A clock label does not identify a unique instant.

06

# DimBalancingAuthority

Logical and physical specification | Conformed dimension

**Grain: One selected balancing authority code.**

| Column | PostgreSQL type | Key / nullability |
|---|---|---|
| **AuthorityKey** | bigint | PK / NOT NULL |
| **AuthorityCode** | text | UQ / NOT NULL |
| **AuthorityName** | text | Attribute / NOT NULL |
| **RegionCode** | text | Attribute / NOT NULL |
| **RegionName** | text | Attribute / NOT NULL |

**Unique natural/business key: AuthorityCode.**

History / update policy: Type 1 for current reference labels. Historical renaming is not reconstructed from a current reference workbook.

06

# DimEnergySource

Logical and physical specification | Conformed dimension

**Grain: One distinct reported fuel column meaning found in the acquired files.**

| Column | PostgreSQL type | Key / nullability |
|---|---|---|
| **EnergySourceKey** | bigint | PK / NOT NULL |
| **SourceColumn** | text | UQ / NOT NULL |
| **SourceLabel** | text | Attribute / NOT NULL |
| **SourceGroup** | text | Attribute / NOT NULL |

**Unique natural/business key: SourceColumn.**

History / update policy: Type 0 meaning. A newly split source category receives a new member. Keep legacy combined categories explicit.

# DimSourceSnapshot

Logical and physical specification | Lineage dimension

### Grain: One retrieved version of one official bulk CSV.

| Column | PostgreSQL type | Key / nullability |
|---|---|---|
| **SourceSnapshotKey** | bigint | PK / NOT NULL |
| **FileName** | text | Attribute / NOT NULL |
| **SourceURL** | text | Attribute / NOT NULL |
| **SHA256** | char(64) | Attribute / NOT NULL |
| **RetrievedAtUTC** | timestamptz | Attribute / NOT NULL |
| **SourceRowCount** | integer | Attribute / NOT NULL |

**Unique natural/business key: (FileName, SHA256).**

History / update policy: Immutable. A different file hash creates a new member; unchanged bytes reuse the member.

# FactHourlyDemand

Logical and physical specification | Hourly Periodic Snapshot Fact

**Grain: One selected authority and one UTC hour-ending instant with its demand, forecast and total net generation.**

| Column | PostgreSQL type | Key / nullability |
|---|---|---|
| **DemandKey** | bigint | PK / NOT NULL |
| **AuthorityKey** | bigint | FK / NOT NULL |
| **DateKey** | bigint | FK / NOT NULL |
| **HourKey** | bigint | FK / NOT NULL |
| **UTCHourEnd** | timestamptz | Attribute / NOT NULL |
| **LocalHourEnd** | timestamp | Attribute / NOT NULL |
| **UTCOffsetMinutes** | smallint | Attribute / NOT NULL |
| **ActualDemandMW** | numeric(18,4) | Attribute / NULL allowed |
| **ForecastDemandMW** | numeric(18,4) | Attribute / NULL allowed |
| **TotalGenerationMW** | numeric(18,4) | Attribute / NULL allowed |
| **SourceSnapshotKey** | bigint | FK / NOT NULL |
| **SourceRowNumber** | integer | Attribute / NOT NULL |

**Unique natural/business key: (AuthorityKey, UTCHourEnd).**

| Foreign key | References (1 : many) |
|---|---|
| **AuthorityKey** | DimBalancingAuthority.AuthorityKey |
| **DateKey** | DimDate.DateKey |
| **HourKey** | DimHour.HourKey |
| **SourceSnapshotKey** | DimSourceSnapshot.SourceSnapshotKey |

History / update policy: Latest published values per business grain. Revisions update the same fact row; preserve raw snapshots for audit.

06

# FactHourlyGeneration

Logical and physical specification | Hourly Periodic Snapshot Fact

**Grain: One selected authority, UTC hour-ending instant and original fuel category with a numeric reported value.**

| Column | PostgreSQL type | Key / nullability |
|---|---|---|
| **GenerationKey** | bigint | PK / NOT NULL |
| **AuthorityKey** | bigint | FK / NOT NULL |
| **DateKey** | bigint | FK / NOT NULL |
| **HourKey** | bigint | FK / NOT NULL |
| **EnergySourceKey** | bigint | FK / NOT NULL |
| **UTCHourEnd** | timestamptz | Attribute / NOT NULL |
| **LocalHourEnd** | timestamp | Attribute / NOT NULL |
| **UTCOffsetMinutes** | smallint | Attribute / NOT NULL |
| **GenerationMW** | numeric(18,4) | Attribute / NOT NULL |
| **SourceSnapshotKey** | bigint | FK / NOT NULL |
| **SourceRowNumber** | integer | Attribute / NOT NULL |

**Unique natural/business key: (AuthorityKey, UTCHourEnd, EnergySourceKey).**

| Foreign key | References (1 : many) |
|---|---|
| **AuthorityKey** | DimBalancingAuthority.AuthorityKey |
| **DateKey** | DimDate.DateKey |
| **HourKey** | DimHour.HourKey |
| **EnergySourceKey** | DimEnergySource.EnergySourceKey |
| **SourceSnapshotKey** | DimSourceSnapshot.SourceSnapshotKey |

History / update policy: Latest published values per business grain. Revisions update the same fact row; preserve raw snapshots for audit.

06

# Physical design rules

Design contract; no database is deployed in Phase 1

| Rule | Specification |
|---|---|
| **Primary keys** | bigint identity surrogate keys, generated once. Preserve on reload; no business meaning in the value. |
| **Uniqueness** | Enforce each table’s natural/business key with a UNIQUE constraint. Composite keys appear explicitly on the table pages. |
| **Referential integrity** | Foreign keys reference dimension PKs. Reject unresolved members into an exception queue. Do not fabricate a normal business member. |
| **Numbers** | numeric(18,4) for all MW measures. Null demand measures remain null. Generation facts exist only for numeric reported values; signed generation is valid. |
| **Time** | UTC end uses timestamptz; local wall-clock end uses timestamp without time zone. UTCOffsetMinutes stores source local minus UTC minutes. |
| **Checks** | Date components follow their dictionary ranges. Authority code and fuel meaning are nonempty. Row numbers are positive. No nonnegative check on signed generation. |
| **Indexes** | Unique business-key indexes support reloads. Add B-tree indexes on fact DateKey and AuthorityKey for time/area filtering; review other indexes against actual queries in Phase 2. |
| **Delete policy** | Restrict deleting referenced dimensions. Controlled partition replacement handles revisions, including withdrawn source values; preserve archived extracts. |
| **Hierarchy** | Date: Year > Quarter > Month > Date; ISOWeekYear > ISOWeekNumber > Date separately. Authority: Region > Authority. Fuel: SourceGroup > exact SourceColumn. Hour is a separate cyclic profile. |